ISHAAN MISRA

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*currently pursuing 7th semester IIT-BHU

Objective

Inquisitive, energetic research oriented student skilled in leadership, with strong foundation in maths, programming logic and cross platform coding and interest in applying new technologies to advance the chemical research process. Seeking to expand my scientific knowledge in the growing field of computational chemical science and to leverage solid development skills with a focus on collaboration, communication, passion, and creativity as a researcher.

Academic Background

Degree	Discipline & Grade	Institution	Country	Date
Integrated Dual Degree (IDD) B. Tech +M. Tech	Industrial Chemistry CPI till 6th semester – 9.48 /10	Indian Institute of Technology (IIT- BHU), Varanasi	India	2020 – Present (Expected 2025)
12th (Senior Secondary Examination)	Science 95 %	Amrita Vidyalayam, New Delhi	India	2020
10th (Secondary Examination)	High School 97%	Amrita Vidyalayam, New Delhi	India	2018

Industrial Experience (Position – As Intern)

May'23-July'23 — **Technische Universität Darmstadt - DAAD WISE Summer Intern**

- Studying computational techniques like all-atomistic molecular dynamics and coarse-grained modelling to create programs for analysing the structural and dynamical properties for simple hydrocarbon molecules.

May-Aug'22 — **Defence Research & Development Organisation(DRDO) (Research & Development Establishment- Engineers)**

- Working on physical and chemical properties of carbon fiber reinforced epoxy resin interleaved composites

Oct-Nov'21 — **Exposys Data Labs, Bengaluru -Full Stack Developer**

- Worked as a web developer and contributed in creating a video-chatting app using react framework and socket IO

July-Dec'21 — **Exploratory Project *(GUIDE: Prof. V. Ramanathan)***

- Simulating differential equation like Lotka-Volterra equation and iodine clock reaction for chemical application(Web-app can be viewed here: https://nonlineardiffeqnsim.netlify.app)
- Exploring the use of logistic maps in chemistry

Publications

2022 — **Published Research Article : BELOUSOV-ZHABOTINSKY REACTION: AN OPEN-SOURCE APPROACH**
DOI: https://doi.org/10.1007/s43538-022-00081-6
Guide: Prof. V Ramanathan

Technical Projects

Mar-Apr'22	Created a full stack picture sharing social media web application using reactjs and tailwind CSS for frontend and sanity for backend database. The web-app can be viewed on: https://shareme-clone-ism.netlify.app
Jan-July'22	Studying modelling cellular competitive growth pattern using python as a programming language
Jan- May'21	Created a simple web application using the react framework and rest APIs that can be used to find the most recent Covid-19 data for different countries. The app can be viewed on the following link: https://ishaanms.github.io/covid_tracking_app/

Area of Expertise

Atomic Modelling & Calculation Software	LAMMPS, VOTCA, Moltemplate, ChemDraw (basic), BURAI (basic), Quantum Espresso (basic)
Programming Languages	Python along with NumPy & Pandas libraries, C++ ,Javascript
Web Technologies	SQL (Basic & Intermediate), MongoDB(Basic & Intermediate), HTML, CSS, Reactjs, Nodejs, Nextjs, Redux

Achievements

- Department Rank 1 at Department of Industrial Chemistry
- Obtained an A* grade in "**Classical Thermodynamics**" course(3rd semester) and "**Quantum chemistry, Group Theory, and Molecular Spectroscopy**" course(4th semester)
- 4 star Coder at Codechef
- HackerRank Badges (https://www.hackerrank.com/ishaanmsr)

C++	★★★★★
Python	★★★★★
Problem Solving	★★★★
SQL	★★★

Additional Information

2021-22	>Member/ Mentor in "Desh Ke Mentor" a Govt of Delhi initiative
2022-23	>Member of Tech team of Science and Technology Council, (SnTC, IIT-BHU)
2022-23	>Member of the Student Alumni Interaction Cell, (SAIC, IIT-BHU)
2018-2019	>Headboy of School "Amrita Vidyalayam New Delhi"
	>Participated in "Ayudh National Leader Summit" – Amritapuri - Kerala
2006-2014	>Have undergone 8 years of formal training in "Tabla"
	>Hobbies include web development, listening to instrumental music, reading sci-fi novels, playing badminton

References

Prof. Dr. Venkatnarayan Ramanathan
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Prof. Dr. Yogesh C Sharma
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